

Nonlinear Trajectory Optimization for Swing-Up of a Double Inverted Pendulum on a Cart

JunYoung Kim

Department of Mechanical and Aerospace Engineering

MAE589: Advanced Robotics Course Project Report

Abstract—This paper studies finite-horizon nonlinear trajectory optimization for swing-up of a double inverted pendulum on a cart. The system is modeled as a six-state, single-input underactuated mechanism with two point-mass links attached to a translating cart, and the swing-up task is posed as a constrained optimal-control problem over the full nonlinear dynamics. The paper first introduces a general optimal-control problem statement, then specializes it to the underactuated nonlinear cart-double-pendulum system. Pontryagin’s minimum principle and the Hamilton-Jacobi-Bellman value-function viewpoint are used to interpret the resulting swing-up formulation before the finite-dimensional optimization problem and simulation results are presented.

Index Terms—trajectory optimization, Pontryagin minimum principle, Hamilton-Jacobi-Bellman equation, direct multiple shooting, nonlinear optimal control, double inverted pendulum

I. Introduction

The double inverted pendulum on a cart is a representative underactuated system in which one horizontal actuator must generate coordinated motion across multiple unstable degrees of freedom. In such systems, the control objective is not simply to reject small perturbations around an equilibrium. Instead, the controller or planner must deliberately create large transient motion, exploit nonlinear coupling, and steer the system into a stabilizable neighborhood of the upright configuration. This is precisely the type of problem for which nonlinear trajectory optimization is attractive.

The swing-up task is especially challenging because the same nonlinear mechanism can exhibit highly sensitive transient motion. Even in passive motion, small differences in phase and angular velocity can produce complicated tip traces and rapid changes in configuration. This sensitivity does not by itself prove mathematical chaos for every parameter setting, but it does illustrate why a purely local controller or a quasi-static plan is not adequate for the global maneuver.

The goal of this paper is to present the trajectory-optimization formulation as a standalone optimal-control study. The emphasis is therefore on the state convention, manipulator-form dynamics, finite-horizon problem statement, Pontryagin minimum principle (PMP), the Hamilton–Jacobi–Bellman (HJB) value-function viewpoint, objective design, and interpretation of the resulting trajectory.

II. Preliminaries and Problem Statement

A. Finite-Horizon Optimal Control

Consider a nonlinear system

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), u(t)), \quad (1)$$

with state $\mathbf{x}(t) \in \mathbb{R}^n$ and input $u(t) \in \mathcal{U}$. A general finite-horizon optimal-control problem can be expressed as

$$\min_{\mathbf{x}(\cdot), u(\cdot)} \int_0^T \ell(\mathbf{x}(t), u(t), t) dt + \phi(\mathbf{x}(T)) \quad (2)$$

$$\text{s.t. } \dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), u(t)), \quad (3)$$

$$\mathbf{x}(0) = \mathbf{x}_{\text{init}}, \quad (4)$$

$$u(t) \in \mathcal{U}. \quad (5)$$

Here ℓ penalizes the path taken during the maneuver, while ϕ penalizes the terminal state. In the cart-double-pendulum swing-up problem, the admissible input set is the actuator interval $\mathcal{U} = [-u_{\text{max}}, u_{\text{max}}]$, and the goal is to connect an initial state near the hanging configuration to a target state $\mathbf{x}^*$ near the upright equilibrium.

B. Underactuation and Nonlinear Behavior

The cart-double-pendulum has three generalized coordinates but only one control input. Thus the system is underactuated: the cart force acts directly on the translational coordinate, while the two angular coordinates move through inertial and gravitational coupling. The swing-up task is therefore a nonlinear planning problem rather than independent tracking of each coordinate.

The nonlinearity is not only a modeling detail; it changes the qualitative behavior of the system. Passive double-pendulum motion is a standard example of sensitive nonlinear dynamics, and the cart-mounted system can produce visually complex tip traces even under zero input, as shown in Fig. 1. This motivates a trajectory-optimization formulation that preserves the full nonlinear dynamics instead of relying only on a local linearization near the upright equilibrium.

III. System Model

A. Coordinates, State, and Control

Let the generalized coordinates be

$$\mathbf{q} = [x \quad \theta_1 \quad \theta_2]^\top, \quad (6)$$

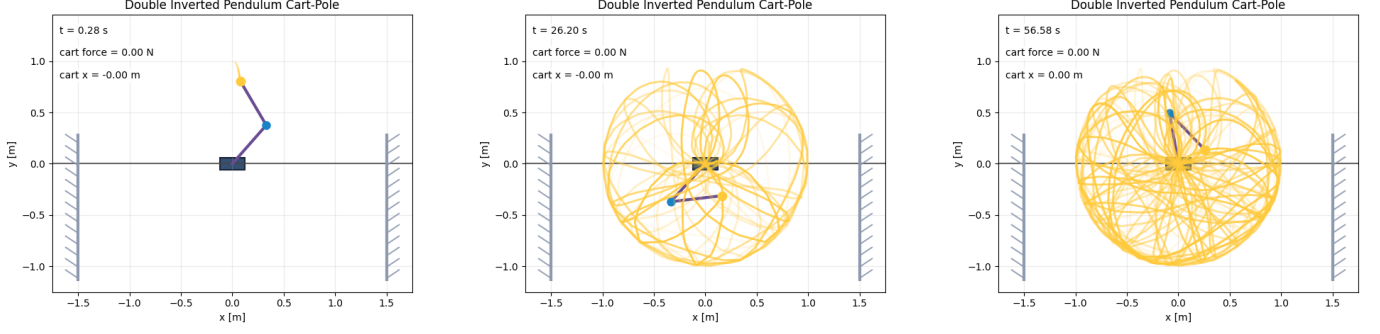


Fig. 1. Representative passive zero-input motion of the cart–double-pendulum at three times from the same rollout. The growing end-tip trace illustrates the complicated transient behavior produced by the nonlinear coupled dynamics even without active control.

where x is the cart position and θ_1, θ_2 are the first and second link angles measured from the upright direction. Thus the upright equilibrium is at

$$\theta_1 = \theta_2 = 0, \quad (7)$$

while a hanging configuration is near

$$\theta_1 = \theta_2 = \pi. \quad (8)$$

The complete state vector is

$$\mathbf{x} = [x \quad \theta_1 \quad \theta_2 \quad \dot{x} \quad \dot{\theta}_1 \quad \dot{\theta}_2]^\top \in \mathbb{R}^6, \quad (9)$$

and the scalar control input is the horizontal cart force

$$u \in \mathbb{R}. \quad (10)$$

The parameter values used for the representative study are

$$m_0 = 1.0 \text{ kg}, \quad m_1 = 0.2 \text{ kg}, \quad m_2 = 0.2 \text{ kg}, \quad (11)$$

$$l_1 = 0.5 \text{ m}, \quad l_2 = 0.5 \text{ m}, \quad g = 9.81 \text{ m/s}^2, \quad (12)$$

with viscous damping $d_x = 0.1$, $d_1 = d_2 = 0.02$, and force limit $u_{\max} = 30 \text{ N}$.

B. Kinematic Convention

Each pendulum link is modeled as a rigid, massless rod with a point mass at its tip. The cart position is

$$\mathbf{p}_c = \begin{bmatrix} x \\ 0 \end{bmatrix}, \quad (13)$$

the first tip position is

$$\mathbf{p}_1 = \mathbf{p}_c + \begin{bmatrix} l_1 \sin \theta_1 \\ l_1 \cos \theta_1 \end{bmatrix}, \quad (14)$$

and the second tip position is

$$\mathbf{p}_2 = \mathbf{p}_1 + \begin{bmatrix} l_2 \sin \theta_2 \\ l_2 \cos \theta_2 \end{bmatrix}. \quad (15)$$

This global-angle representation fixes the angular convention used throughout the formulation and motivates periodic angular costs rather than raw Euclidean angle differences.

C. Dynamic Model

The dynamics take the manipulator form

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} = \boldsymbol{\tau}(\mathbf{q}, \dot{\mathbf{q}}, u), \quad (16)$$

where $\mathbf{M}(\mathbf{q})$ is the configuration-dependent mass matrix, $\ddot{\mathbf{q}}$ is the generalized acceleration vector, and $\boldsymbol{\tau}(\mathbf{q}, \dot{\mathbf{q}}, u)$ collects all generalized forces and torques, including control input, damping, gravity, and velocity-coupling terms. Writing the model in this form separates inertia from the remaining forces. The mass matrix is

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} m_0 + m_1 + m_2 & (m_1 + m_2)l_1 \cos \theta_1 & m_2 l_2 \cos \theta_2 \\ (m_1 + m_2)l_1 \cos \theta_1 & (m_1 + m_2)l_1^2 & m_2 l_1 l_2 \cos(\theta_1 - \theta_2) \\ m_2 l_2 \cos \theta_2 & m_2 l_1 l_2 \cos(\theta_1 - \theta_2) & m_2 l_2^2 \end{bmatrix} \quad (17)$$

and the generalized-force vector is

$$\boldsymbol{\tau} = \begin{bmatrix} u - d_x \dot{x} + (m_1 + m_2)l_1 \sin \theta_1 \dot{\theta}_1^2 + m_2 l_2 \sin \theta_2 \dot{\theta}_2^2 \\ -d_1 \dot{\theta}_1 - m_2 l_1 l_2 \sin(\theta_1 - \theta_2) \dot{\theta}_2^2 + (m_1 + m_2)g l_1 \sin \theta_1 \\ -d_2 \dot{\theta}_2 + m_2 l_1 l_2 \sin(\theta_1 - \theta_2) \dot{\theta}_1^2 + m_2 g l_2 \sin \theta_2 \end{bmatrix}. \quad (18)$$

The off-diagonal terms of $\mathbf{M}(\mathbf{q})$ encode the coupling between cart motion and link motion, as well as the coupling between the two pendulum links. This coupling is central to the swing-up maneuver because the angular coordinates are not directly actuated.

Solving (16) yields $\ddot{\mathbf{q}}$, and the first-order state dynamics become

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u) = [\dot{x} \quad \dot{\theta}_1 \quad \dot{\theta}_2 \quad \ddot{x} \quad \ddot{\theta}_1 \quad \ddot{\theta}_2]^\top. \quad (19)$$

D. Task Definition

The representative swing-up task sets

$$\mathbf{x}_0 = [0 \quad \pi + 0.08 \quad \pi - 0.08 \quad 0 \quad 0 \quad 0]^\top \quad (20)$$

as the initial state and

$$\mathbf{x}^* = [3 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^\top \quad (21)$$

as the target state. The discretization uses $N = 40$ multiple-shooting intervals with $\Delta t = 0.08 \text{ s}$, corresponding to a planning horizon of 3.2 s.

IV. Optimal-Control Interpretation

A. Pontryagin Minimum Principle

Before discretizing the problem, it is useful to interpret the swing-up task as a continuous-time constrained optimal-control problem. Let $L(\mathbf{x}, u, t)$ denote the running cost and $\phi(\mathbf{x}(T))$ denote the terminal cost in (5). Following the sign convention used in class, introduce the costate $\mathbf{p}(t) \in \mathbb{R}^6$ and define the Hamiltonian

$$H(\mathbf{x}, u, \mathbf{p}, t) = \mathbf{p}^\top \mathbf{f}(\mathbf{x}, u) - L(\mathbf{x}, u, t). \quad (22)$$

The costate $\mathbf{p}$ acts as a Lagrange multiplier for the differential constraint $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u)$. Equivalently, the constrained functional can be written as

$$\bar{J} = \phi(\mathbf{x}(T)) + \int_0^T [L(\mathbf{x}, u, t) + \mathbf{p}^\top (\mathbf{f}(\mathbf{x}, u) - \dot{\mathbf{x}})] dt. \quad (23)$$

Taking the first variation of (23) yields the standard PMP necessary conditions:

$$\dot{\mathbf{x}}^*(t) = \frac{\partial H}{\partial \mathbf{p}}(\mathbf{x}^*(t), u^*(t), \mathbf{p}^*(t), t), \quad (24)$$

$$\dot{\mathbf{p}}^*(t) = -\frac{\partial H}{\partial \mathbf{x}}(\mathbf{x}^*(t), u^*(t), \mathbf{p}^*(t), t), \quad (25)$$

$$0 = \frac{\partial H}{\partial u}(\mathbf{x}^*(t), u^*(t), \mathbf{p}^*(t), t), \quad (26)$$

$$\mathbf{p}^*(T) = -\nabla_{\mathbf{x}} \phi(\mathbf{x}^*(T)). \quad (27)$$

The stationarity condition (26) is the interior-control form; with an active actuator bound, it is interpreted through the corresponding constrained maximization over $\mathcal{U}$. These equations form a two-point boundary-value problem, with the state fixed initially and the costate tied to the terminal cost.

B. Hamilton–Jacobi–Bellman Viewpoint

The same problem can also be viewed through dynamic programming. Whereas PMP describes necessary conditions along one optimal trajectory, HJB describes the optimal cost-to-go from arbitrary states. Define the value function

$$V(\mathbf{x}, t) = \min_{u(\cdot)} \left[\phi(\mathbf{x}(T)) + \int_t^T L(\mathbf{x}(\tau), u(\tau), \tau) d\tau \right], \quad (28)$$

where the minimization is over admissible controls applied from time t to T , with the trajectory initialized at the state $\mathbf{x}$ at time t . Bellman's principle of optimality gives the HJB equation

$$\frac{\partial V}{\partial t}(\mathbf{x}, t) = \sup_{u \in \mathcal{U}} \left[-L(\mathbf{x}, u, t) - \nabla_{\mathbf{x}} V(\mathbf{x}, t)^\top \mathbf{f}(\mathbf{x}, u) \right], \quad (29)$$

The terminal cost in the finite-horizon problem becomes the boundary condition

$$V(\mathbf{x}, T) = \phi(\mathbf{x}). \quad (30)$$

With the sign convention in (22), the expression maximized in (29) is the Hamiltonian in the PMP condition when

$$\mathbf{p}(t) = -\nabla_{\mathbf{x}} V(\mathbf{x}(t), t). \quad (31)$$

Thus PMP and HJB are complementary interpretations of the same optimal-control problem. PMP uses the costate as an adjoint sensitivity along the optimized path, while HJB stores the corresponding sensitivity as the gradient of a value function over state space. For the present project, solving the full nonlinear HJB partial differential equation is not practical, but the viewpoint is still useful: it clarifies why terminal penalties approximate future cost-to-go near the upright equilibrium and why a local quadratic value approximation would naturally lead to Riccati/LQR-type stabilization after the swing-up plan.

V. Discrete Optimal-Control Formulation

A. Reference Curve

A smooth reference curve is used as a nominal path for the stage cost and initialization. Let

$$\tau_k = \frac{k}{N}, \quad k = 0, \dots, N, \quad (32)$$

and define the cubic smoothstep profile

$$s(\tau) = 3\tau^2 - 2\tau^3. \quad (33)$$

Then the initialization reference is

$$x_k^{\text{ref}} = x_0 + (x^* - x_0)s(\tau_k), \quad (34)$$

$$\theta_{1,k}^{\text{ref}} = \theta_{1,0} + (\theta_1^* - \theta_{1,0})s(\tau_k), \quad (35)$$

$$\theta_{2,k}^{\text{ref}} = \theta_{2,0} + (\theta_2^* - \theta_{2,0})s(\tau_k), \quad (36)$$

while the velocity components of the reference are set to zero:

$$\dot{x}_k^{\text{ref}} = \dot{\theta}_{1,k}^{\text{ref}} = \dot{\theta}_{2,k}^{\text{ref}} = 0. \quad (37)$$

This curve is not required to satisfy the dynamics; dynamic feasibility is imposed separately by the shooting constraints.

B. Nominal Input Reference

For each reference state, a feedforward force is chosen by minimizing the squared generalized acceleration:

$$u_k^{\text{ref}} \in \arg \min_{u \in \mathcal{U}} \|\ddot{\mathbf{q}}(\mathbf{x}_k^{\text{ref}}, u)\|_2^2. \quad (38)$$

Thus (38) gives a dynamically interpretable nominal input for the cost function without replacing the nonlinear dynamic constraints.

C. Decision Variables

After time discretization, the nonlinear program uses the state and input decision variables

$$\mathbf{X} = [\mathbf{x}_0 \quad \mathbf{x}_1 \quad \cdots \quad \mathbf{x}_N] \in \mathbb{R}^{6 \times (N+1)} \quad (39)$$

and

$$\mathbf{U} = [u_0 \quad u_1 \quad \cdots \quad u_{N-1}] \in \mathbb{R}^{1 \times N}. \quad (40)$$

Although $\mathbf{x}_0$ is part of the variable matrix, it is constrained to equal the prescribed initial state:

$$\mathbf{x}_0 = \mathbf{x}_{\text{init}}. \quad (41)$$

D. RK4 Defect Constraints

Dynamic feasibility is enforced through direct multiple shooting. For each stage $k = 0, \dots, N-1$, the next state variable is constrained to equal one RK4 step of the symbolic dynamics:

$$\mathbf{x}_{k+1} = \Phi_{\text{RK4}}(\mathbf{x}_k, u_k), \quad (42)$$

where Φ_{RK4} is defined by

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{x}_k, u_k), \quad (43)$$

$$\mathbf{k}_2 = \mathbf{f}\left(\mathbf{x}_k + \frac{\Delta t}{2} \mathbf{k}_1, u_k\right), \quad (44)$$

$$\mathbf{k}_3 = \mathbf{f}\left(\mathbf{x}_k + \frac{\Delta t}{2} \mathbf{k}_2, u_k\right), \quad (45)$$

$$\mathbf{k}_4 = \mathbf{f}(\mathbf{x}_k + \Delta t \mathbf{k}_3, u_k), \quad (46)$$

$$\Phi_{\text{RK4}}(\mathbf{x}_k, u_k) = \mathbf{x}_k + \frac{\Delta t}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \quad (47)$$

The shooting defects are the discrete counterpart of the state equation in the PMP and HJB formulations.

E. Control Constraint

The optimizer imposes the control bounds

$$-u_{\max} \leq u_k \leq u_{\max}, \quad k = 0, \dots, N-1. \quad (48)$$

In the nominal swing-up formulation, no obstacle, link-span, or state-rate constraints are included. The obstacle-aware experiment below augments this nominal problem with a soft workspace collision penalty while keeping the same nonlinear dynamics and direct-transcription structure.

F. Obstacle-Aware Extension

Let $\mathcal{P}(\mathbf{x}_k)$ denote a finite set of sampled points on the cart-double-pendulum geometry at time step k . In the implementation, this set contains the cart position and uniformly spaced sample points along both links. For a rectangular obstacle o with center $\mathbf{c}_o = (c_x, c_y)$, half-width a_o , half-height b_o , and clearance margin r_{safe} , define the inflated elliptical intrusion field

$$\rho_o(\mathbf{p}) = 1 - \left(\frac{p_x - c_x}{a_o + r_{\text{safe}}} \right)^2 - \left(\frac{p_y - c_y}{b_o + r_{\text{safe}}} \right)^2. \quad (49)$$

Positive ρ_o indicates that the point lies inside the inflated safety field. The obstacle cost is

$$J_{\text{obs}} = \sum_{k=0}^N \sum_{\mathbf{p} \in \mathcal{P}(\mathbf{x}_k)} \sum_{o \in \mathcal{O}} w_{\text{obs}} [\text{softplus}_{\beta}(\rho_o(\mathbf{p}))]^2, \quad (50)$$

where

$$\text{softplus}_{\beta}(z) = \frac{1}{\beta} \log(1 + \exp(\beta z)). \quad (51)$$

This construction leaves the problem as a smooth nonlinear program rather than introducing hard collision constraints. It is therefore easier to solve with IPOPT while still discouraging the optimizer from placing the cart or links near obstacles.

VI. Objective Function

A. Path Cost

The running objective combines quadratic state tracking, periodic angle costs, control deviation penalties, and control smoothness penalties. Let

$$\delta \mathbf{x}_k = \mathbf{x}_k - \mathbf{x}_k^{\text{ref}}. \quad (52)$$

The path-angle penalty uses

$$c_{\theta}(\theta, \theta^{\text{ref}}) = 1 - \cos(\theta - \theta^{\text{ref}}), \quad (53)$$

which respects angular periodicity and avoids the ambiguity of large wrapped rotations. The stage cost is therefore

$$\begin{aligned} \ell_k &= \delta \mathbf{x}_k^{\top} \mathbf{Q}_{\text{path}} \delta \mathbf{x}_k \\ &\quad + w_{\theta,p} c_{\theta}(\theta_{1,k}, \theta_{1,k}^{\text{ref}}) + w_{\theta,p} c_{\theta}(\theta_{2,k}, \theta_{2,k}^{\text{ref}}) \\ &\quad + w_u (u_k - u_k^{\text{ref}})^2 + w_{\Delta u} (u_k - u_{k-1})^2, \end{aligned} \quad (54)$$

where the smoothness term is applied only for $k > 0$. The angular states are weighted through c_{θ} rather than through the diagonal entries of $\mathbf{Q}_{\text{path}}$, preserving the periodic geometry of the configuration variables while retaining quadratic penalties for translation and velocity. The numerical weights used for simulation are collected in Table I.

B. Terminal Cost

The terminal state is penalized using

$$\delta \mathbf{x}_N = \mathbf{x}_N - \mathbf{x}^*, \quad (55)$$

The terminal cost is

$$\begin{aligned} \phi(\mathbf{x}_N) &= \delta \mathbf{x}_N^{\top} \mathbf{Q}_{\text{term}} \delta \mathbf{x}_N \\ &\quad + w_{\theta,t} c_{\theta}(\theta_{1,N}, \theta_1^*) + w_{\theta,t} c_{\theta}(\theta_{2,N}, \theta_2^*), \end{aligned} \quad (56)$$

where

$$w_{\theta,t} > w_{\theta,p}. \quad (57)$$

The terminal cost uses the same geometric separation as the path cost, but places greater emphasis on the final cart position, velocities, and upright angular agreement.

TABLE I
Simulation Parameters and Objective Weights

Quantity	Value
State dimension	6
Input dimension	1
Horizon steps N	40
Time step Δt	0.08 s
Planning horizon	3.2 s
Goal cart position	3.0 m
Initial angle offset	0.08 rad
$\mathbf{Q}_{\text{path}}$	diag(8, 0, 0, 0.35, 0.2, 0.2)
$\mathbf{Q}_{\text{term}}$	diag(65, 0, 0, 5, 3, 3)
$w_{\theta,p}$	10.0
$w_{\theta,t}$	140.0
w_u	0.003
$w_{\Delta u}$	0.0008

C. Total Optimization Problem

Putting all pieces together, the discretized nonlinear program can be summarized as

$$\min_{\mathbf{x}, \mathbf{u}} \sum_{k=0}^{N-1} \ell_k + \phi(\mathbf{x}_N) + J_{\text{obs}} \quad (58)$$

$$\text{s.t. } \mathbf{x}_0 = \mathbf{x}_{\text{init}}, \quad (59)$$

$$\mathbf{x}_{k+1} = \Phi_{\text{RK4}}(\mathbf{x}_k, u_k), \quad k = 0, \dots, N-1, \quad (60)$$

$$-u_{\max} \leq u_k \leq u_{\max}, \quad k = 0, \dots, N-1. \quad (61)$$

VII. Simulation Results

The representative simulation uses the parameter and objective weights in Table I. The result is summarized both by the terminal state and by the trajectory plot in Fig. 3. A planned two-column snapshot visualization is reserved in Fig. 2; this figure is intended to show the full swing-up motion as a sequence of cart–pendulum configurations along the optimized trajectory.

For the parameters above, the optimized terminal state was

$$\mathbf{x}_N \approx \begin{bmatrix} 3.0472 \\ 0.0017 \\ -0.0023 \\ 0.3967 \\ 0.3514 \\ 0.2686 \end{bmatrix}^{\top}, \quad (62)$$

relative to the target

$$\mathbf{x}^* = [3 \ 0 \ 0 \ 0 \ 0 \ 0]^{\top}. \quad (63)$$

Thus the optimized plan reaches the desired upright vicinity quite accurately in position and angle, while still carrying noticeable terminal velocities. This is consistent with the structure of the objective: the terminal state is strongly penalized, but not constrained to match the goal exactly.

The obstacle-aware run used the same 3.2 s horizon with $N = 80$, two rectangular workspace obstacles, $w_{\text{obs}} =$

1600, and $r_{\text{safe}} = 0.16$ m. The resulting terminal state was approximately

$$[3.0552 \ 0.0246 \ 0.0201 \ 0.4414 \ 0.3740 \ 0.2416]^{\top}, \quad (64)$$

with peak control magnitude 49.996 N and minimum sampled obstacle clearance 0.0998 m. This demonstrates that the same trajectory-optimization framework can generate a swing-up maneuver that avoids workspace obstacles without relying on the final LQR handoff.

Because the terminal velocities are nonzero, the trajectory should be interpreted as a finite-horizon swing-up motion that reaches the upright neighborhood, not as a proof of asymptotic stabilization.

VIII. Discussion and Extensions

The trajectory-optimization formulation answers a direct finite-horizon planning question: what state and input sequence best connects the hanging configuration to the upright target under the full nonlinear dynamics? The absence of an invariant terminal set or feedback law means the result should be interpreted as a nominal open-loop extremal candidate, consistent with the PMP/HJB formulation, rather than as a complete closed-loop stabilizing controller.

IX. Conclusion

This paper documented a nonlinear trajectory-optimization formulation for swing-up of a double inverted pendulum on a cart. The formulation combines a six-state manipulator-form model, a finite-horizon PMP/HJB interpretation, nonlinear dynamic feasibility constraints, and a shaped path-and-terminal objective.

The main conceptual takeaway is that the swing-up result depends on the structure of the optimal-control problem: periodic angular costs, effort regularization, terminal weighting, and nonlinear dynamic defects jointly shape the feasible maneuver. The resulting trajectory reaches the upright neighborhood with small position and angular terminal errors, though not yet a strict rest-to-rest endpoint.

References

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Full-horizon trajectory snapshot figure placeholder.

Fig. 2. Placeholder for planned full-horizon trajectory snapshots of the cart–double-pendulum swing-up motion.

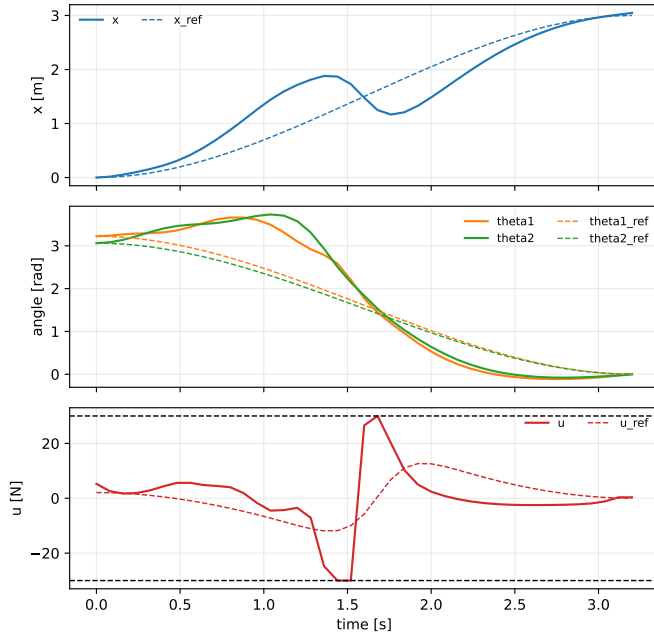


Fig. 3. Optimized state and control trajectory. Solid lines show optimized states and control; dashed lines show the reference curve and feedforward input.